

[DC] Supporting Embodied Sensemaking in Immersive Environment

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ABSTRACT

Traditional desktop-based visualisations have their limitations in supporting and improving the sensemaking of data that is inherent 3D, such as spatio-temporal data. Immersive analytics and embodiment provide new opportunities to understand this kind of data in a more engaging way. However, little research explored how to support the whole sensemaking process in immersive environment, from low level of data understanding and exploration to higher level of provenance. My doctoral research focuses on supporting spatio-temporal data sensemaking using embodiment in immersive environment. We hope to provide guidance for developing embodied tools that aim to accomplish various levels of sensemaking tasks in immersive spaces.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Visualization

1 INTRODUCTION AND RELATED WORK

Currently, data visualisation, as with most interactive computing tasks, is overwhelmingly performed using flat screens. However, for spatio-temporal data which is inherent 3D and typically visualised with a map representation to display spatial attributes [17], this presents a problem, as rendering 3D data displays on screens is well known to suffer from issues of occlusion, perspective distortion and so on, and generally a loss of information [11]. With the advancement and affordability of immersive technologies such as Augmented Reality (AR) and Virtual Reality (VR), immersive analytics [9] has brought rich opportunities for the exploration of data visualisations.

A compelling aspect of immersive analytics is embodiment, the mapping of data artifacts to 3D virtual constructs that users can directly manipulate, examine, and rearrange [4]. It has been used in recent research to support data exploration. For example, ImAxes [2] allows users to construct ad-hoc visualisations by combining individual axes. Recent works also take advantage of large spatial environments afforded by Virtual Reality (VR) for organising and browsing embodied data constructs [5]. The ability to support intuitive interaction and embodied data representations makes it beneficial for supporting data sensemaking in a natural and engaging manner.

Sensemaking is the process of extracting information and building understanding from a collection and organisation of data [1, 6]. Pirolli and Card [12] describe sensemaking as an iterative process that contains two main loops: foraging and sensemaking, during which raw data is transformed into rational explanation through evidence extraction, schema organization and hypothesis generation. Many existing immersive analytics applications aim to support sensemaking through abstract data visualizations. However, researchers in this field have not defined explicit connections between concepts encompassed by embodied interaction and spatial user interface design and Pirolli and Card's sensemaking model. For instance, while many interactive tasks that current immersive analytics tools support, such as selecting and filtering data [3, 7], connecting and organising

multiple views [6, 13], are compartmentalised by this model, it does not describe how inherent cognitive abilities such as spatial memory, kinaesthesia and proprioception [8, 10] can be directly applied to support analytic tasks.

In addition, most previous work concentrated on stages in the foraging loop of Pirolli and Card's sensemaking model, such as data specification and view manipulation. More recently, researchers have come to realise that collecting and reflecting on *analytic provenance* information — knowledge about the history of users' interactions and reasoning through this complex reasoning process [16] — is in itself an important part of sensemaking [14, 15]. However, most research on analytic provenance for sensemaking is browser-based, to our knowledge, no study has systematically investigated how analytic provenance can be represented and used for supporting sensemaking tasks with embodiment in immersive environment.

In this doctoral thesis, we focus on filling this gap by using embodied data constructs, embodied interactions and large space organisation to support the sensemaking process of spatio-temporal data, especially including provenance tasks in the sensemaking loop, and offering recommendations and guidelines for such embodied analysis tools.

2 CURRENT PROGRESS

In this research, we break the sensemaking process into three stages from low level to high level: 1) data understanding, 2) data exploration, and 3) analytic provenance. We then plan to work on one project for each stage.

Our first project focuses on the perception phrase of the embodied data representation. We worked on exploring the effectiveness of different design of data representation, such as encoding, scale, etc, on understanding the spatio-temporal data. Using a space-time cube visualisation of building data, we provided three scales (a small tabletop scale, a medium scale and a large scale) for the visualisation encoded by three visual encoding (colour hue, area and thickness). We conducted a user study to investigate these variables with 18 participants. We collected a wide range of data including response time and accuracy, subjective measures of workload, simulator sickness and participants' preference. Our results indicate that colour is more accurately perceived than the other two encoding, requires lower workload, and is most preferred by participants. Area is particularly problematic with large scale visualisations, and a 'medium' scale, which places the visualisation at eye level, is most highly preferred.

From participants' feedback in this user study, we also identified building shapes and number of buildings as two interesting variables and two user studies for evaluating how they affect the perception of encoding for data understanding are working in progress.

To support data exploration, more interactions are required in addition to the perception of data representations. Therefore, we moved on to the embodied interactions in our second project. We identified several design criteria for a system that supports embodied exploration of data over space and time. We implemented a prototype system called TimeTables to support data exploration, using embodiment with space-time cubes in virtual reality (see Figure 1). TimeTables uses multiple space-time cubes on virtual tabletops, which users can manipulate by extracting time layers or individual buildings to create new tabletop views. The surrounding environment includes a large space for multiple linked tabletops and a storage wall. TimeTables presents information at different time scales by

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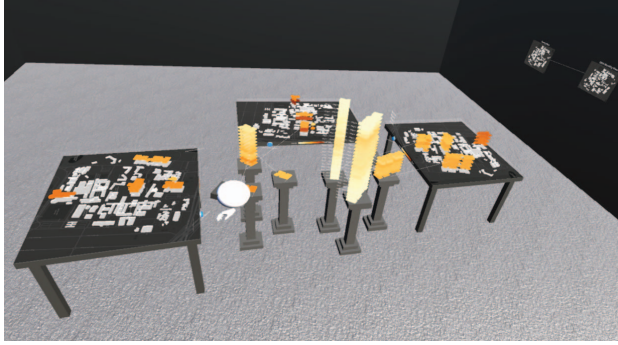


Figure 1: An overview of TimeTables.

stretching layers to drill down in time. From our use case scenario and analysis, we believe the system has a high potential in assisting spatio-temporal data exploration and provide recommendations for similar embodied exploration tools.

In our third project, we focused on the potential of embodiment and immersive space for supporting analytic provenance. We proposed the concept of *embodied provenance*, the use of virtual space and physical interactions in supporting tracking, understanding, and reconstruction of analysis history. We identified a set of design criteria and implemented a prototype system to demonstrate this concept in VR, by providing multiple data views, spatial environments and novel embodied interactions (see Figure 2). We evaluated the system through qualitative evaluation with experts in the domains of provenance, immersive analytics and data visualisation, which shows the potential and promise of the concept and the prototype system. Our exploration of embodied provenance through this prototype provides lessons learnt to guide the design of immersive analytic tools for embodied provenance. More important, this research sheds light on the gaps of how to leverage the benefits of embodied data constructs and interactions to improve the understanding of analysis history to support sensemaking.

3 CONCLUSION

So far, we have explored the use of embodiment in supporting data understanding and exploration of the sensemaking process and our results are positive. In the next step, we will focus on advanced exploration of embodied provenance and hope our continuous work could bring more attention to the analytic provenance in immersive environment and provide guideline in the early stage of this field.

During the doctoral consortium, I welcome any feedback on the current process or the proposed next steps. I am wondering if it would be a good idea to extend the application of the embodied provenance concept in AR in my next steps and if there are any aspects that were not mentioned that should be inquired in the scope of this thesis.

REFERENCES

- [1] C. Andrews, A. Endert, and C. North. Space to think: Large high-resolution displays for sensemaking. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI '10, p. 55–64. Association for Computing Machinery, New York, NY, USA, 2010. doi: 10.1145/1753326.1753336
- [2] M. Cordeil, A. Cunningham, T. Dwyer, B. H. Thomas, and K. Marriott. Imaxes: Immersive axes as embodied affordances for interactive multivariate data visualisation. *UIST '17*, p. 71–83. Association for Computing Machinery, New York, NY, USA, 2017. doi: 10.1145/3126594.3126613
- [3] K. Danyluk, T. T. Ulusoy, W. Wei, and W. Willett. Touch and beyond: Comparing physical and virtual reality visualizations. *IEEE Transactions on Visualization and Computer Graphics*, 2020.

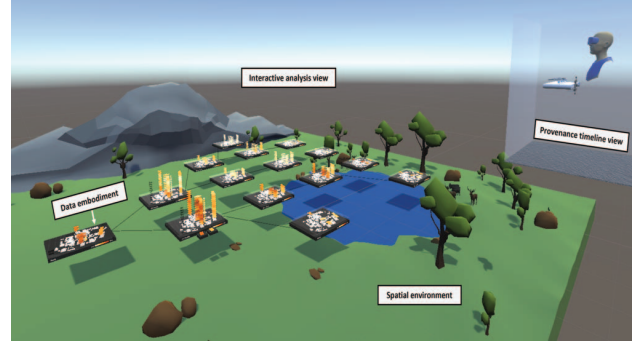


Figure 2: An overview of prototype for embodied provenance.

- [4] P. Dourish. *Where the action is: the foundations of embodied interaction*. MIT press, 2004.
- [5] D. Hayatpur, H. Xia, and D. Wigdor. Datahop: Spatial data exploration in virtual reality. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology*, UIST '20, p. 818–828. Association for Computing Machinery, New York, NY, USA, 2020. doi: 10.1145/3379337.3415878
- [6] L. Lisle, K. Davidson, E. J. Gitre, C. North, and D. A. Bowman. Sensemaking strategies with immersive space to think. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, pp. 529–537, 2021. doi: 10.1109/VR50410.2021.00077
- [7] J. Liu, A. Prouzeau, B. Ens, and T. Dwyer. Design and evaluation of interactive small multiples data visualisation in immersive spaces. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 588–597. IEEE, 2020.
- [8] K. Marriott, J. Chen, M. Hlawatsch, T. Itoh, M. A. Nacenta, G. Reina, and W. Stuerzlinger. *Immersive Analytics: Time to Reconsider the Value of 3D for Information Visualisation*, pp. 25–55. Springer International Publishing, Cham, 2018. doi: 10.1007/978-3-030-01388-2_2
- [9] K. Marriott, F. Schreiber, T. Dwyer, K. Klein, N. H. Riche, T. Itoh, W. Stuerzlinger, and B. H. Thomas. *Immersive Analytics*, vol. 11190. Springer, 2018.
- [10] J. McCormack, J. C. Roberts, B. Bach, C. D. S. Freitas, T. Itoh, C. Hurter, and K. Marriott. *Multisensory Immersive Analytics*, pp. 57–94. Springer International Publishing, Cham, 2018. doi: 10.1007/978-3-030-01388-2_3
- [11] T. Munzner. *Visualization analysis and design*. CRC press, 2014.
- [12] P. Pirolli and S. Card. The sensemaking process and leverage points for analyst technology as identified through cognitive task analysis. In *Proceedings of international conference on intelligence analysis*, vol. 5, 2005.
- [13] K. A. Satriadi, B. Ens, M. Cordeil, T. Czuderna, and B. Jenny. Maps around me: 3d multiview layouts in immersive spaces. *Proceedings of the ACM on Human-Computer Interaction*, 4(ISS):1–20, 2020.
- [14] A. Toniolo, T. J. Norman, A. Etuk, F. Cerutti, R. W. Ouyang, M. Sri-vastava, N. Oren, T. Dropps, J. A. Allen, and P. Sullivan. Supporting reasoning with different types of evidence in intelligence analysis. In *Proceedings of the 2015 International Conference on Autonomous Agents and Multiagent Systems*, AAMAS '15, p. 781–789. International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 2015.
- [15] A. Wexelblat and P. Maes. Footprints: History-rich tools for information foraging. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI '99, p. 270–277. Association for Computing Machinery, New York, NY, USA, 1999. doi: 10.1145/302979.303060
- [16] K. Xu, S. Attfield, T. Jankun-Kelly, A. Wheat, P. H. Nguyen, and N. Selvaraj. Analytic provenance for sensemaking: A research agenda. *IEEE Computer Graphics and Applications*, 35(3):56–64, 2015. doi: 10.1109/MCG.2015.50
- [17] L. Yang, Z. Ma, L. Zhu, and L. Liu. Research on the visualization of spatio-temporal data. In *IOP Conference Series: Earth and Environmental Science*, vol. 234, p. 012013. IOP Publishing, 2019.